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RECURSION VS. ITERATION



Neither recursion nor iteration is a superior technique in general. In fact, any recursive code can be written as iterative code with a loop and a stack. Recursion doesn't have some special power enabling it to perform calculations that an iterative algorithm cannot. And any iterative loop can be rewritten as a recursive function.

This chapter compares and contrasts recursion and iteration. We'll look at the classic Fibonacci and factorial functions and see why their recursive algorithms have critical weaknesses. We'll also explore the insights a recursive approach can yield by considering an exponent algorithm. Altogether this chapter shines light on the supposed elegance of recursive algorithms and shows when a recursive solution is useful and when it is not.